

Course Project Review 1

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A Strong Title Defines the Project

Key idea

The project title should clearly communicate the **main topic, purpose, and scope** of the proposed work.

Example

Quantum-Guided Deep-Learning Framework for the Early Diagnosis and Prognosis of Idiopathic Pulmonary Fibrosis

Objectives State What the Project Will Achieve

Definition

An objective describes the **specific, measurable goal** that the project aims to accomplish.

Example objective

To develop a deep-learning model that accurately detects pulmonary fibrosis from chest CT images.

Gap Analysis Identifies What Is Missing

Gap analysis compares the **current situation** with the **desired situation**. It reveals the limitations that the proposed project should address.

- ▶ Identify weaknesses in the existing system.
- ▶ Define the required improvement.
- ▶ Connect the research problem to the proposed solution.

Core relationship

$$\text{Gap} = \text{Desired State} \\ - \text{Current State}$$

The Proposed System Is the Planned Solution

Definition

The proposed system is the **new method, model, or application** designed to solve the problems identified through gap analysis.

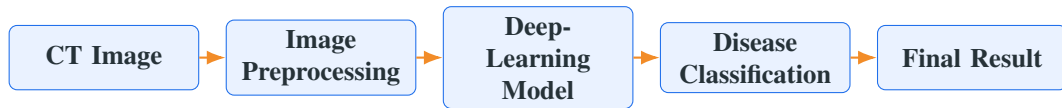
Example

The proposed system uses a deep-learning model to analyse chest CT images and automatically classify them as **normal** or **pulmonary fibrosis**.

Architecture Design Shows Components and Data Flow

Definition

Architecture design presents the overall structure of the project, including its major components, their connections, and the movement of data.



Tools Provide the Project's Technical Foundation

What should be included?

Programming languages, libraries, frameworks, platforms, hardware, and datasets used to develop and evaluate the project.

- ▶ **Python** – programming language
- ▶ **PyTorch** – deep-learning framework
- ▶ **Google Colab** – coding and training platform
- ▶ **OpenCV** – image preprocessing
- ▶ **Dataset** – training and testing data
- ▶ **GPU/CPU** – model-training hardware

Modules Divide the Project into Manageable Parts

Definition

Module identification divides the complete project into smaller components, with each module responsible for a specific task.

- ▶ Data collection module
- ▶ Preprocessing module
- ▶ Classification module
- ▶ Performance-evaluation module
- ▶ Result-display module

Explain Every Module Through Input, Process, and Output

Module explanation

Describe the **purpose**, **input**, **processing steps**, and **output** of every identified module.

Example: Image Preprocessing Module

- ▶ **Input:** Raw chest CT images
- ▶ **Process:** Resizing, normalization, and noise removal
- ▶ **Output:** Clean, standardized images for model training

Choose an Algorithm and Explain How It Works

Algorithm identification

Select the most suitable computational method for solving the project's problem.

Algorithm explanation

Describe how the algorithm processes its input and produces the required output.

Example: Convolutional Neural Network (CNN)

A CNN learns important visual features from chest CT images and uses them to classify each image as normal or pulmonary fibrosis.

The First 25% Should Demonstrate Visible Progress

Expected milestone

Approximately one-fourth of the planned development work should be completed and supported by initial evidence or a working prototype.

- ▶ Project environment and required tools are configured.
- ▶ The dataset is collected, organized, and loaded.
- ▶ Initial preprocessing is completed.
- ▶ Basic modules or a prototype are developed.
- ▶ Preliminary testing has started and early results are recorded.